Graph Traversal and Shortest Paths

LumoraAI Presentation Demo • 01 Data Structures Algorithms

BFS and DFS

• Breadth-first search explores vertices level by level using a queue. In an unweighted graph, the first time a vertex is reached gives a shortest path in number of edges.

• Depth-first search explores as far as possible before backtracking. It is commonly implemented with recursion or an explicit stack and is useful for reachability, cycle analysis and topological-ordering techniques.

Complexity

• With adjacency lists, BFS and DFS both run in O(V + E), where V is the number of vertices and E is the number of edges, because each vertex and edge is processed a bounded number of times.

Dijkstra overview

• Dijkstra's algorithm repeatedly selects the currently known closest unsettled vertex and relaxes its outgoing edges. It assumes non-negative edge weights. With a binary heap and adjacency lists, a common complexity bound is O((V + E) log V).

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